

Born-Compatible Record Measures and the Classical Concentration Limit: Separate Theorems and Their MTT Interface

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Abstract

The Born rule and the emergence of classical behavior are related, but they are not the same mathematical problem. The first fixes outcome probabilities for a declared quantum preparation and instrument. The second asks when retained records and observables are well approximated by one stable classical alternative. We formulate their interface without identifying them. Standard Gleason-type results recover trace probabilities only after additive or effect-noncontextual probability assumptions are supplied; they do not select an MTT source measure. The current q79 program now provides a stronger, domain-specific result: its canonical binary one-anchor Fock recorder emits an exact stopped-output measure and second-moment capture descent from the selected normal state, without an added Born axiom, stochastic primitive, fit, or observed probability on that domain. General apparatus contexts and objective single-history actualization remain open. Separately, we prove exact concentration and persistence bounds. If one record has probability at least $1 - \varepsilon$, the law is within ε in total variation of a deterministic record, bounded observables differ by at most ε times their oscillation, and a record with per-step escape probability at most η survives n steps with probability at least $1 - \varepsilon - n\eta$. These statements define a controlled classical limit. They do not derive the outcome weights or select one realized history.

Version 3 Revision Note

Supersedes Version 2 as the current manuscript. The released identity and previous revision note are preserved.

Reason The canonical output-measure result needs its source chain, state update, and conditional extension hypotheses explained alongside the separate concentration theorems.

Resolution Five frozen sources are integrated: output measure, minimal recorder action, Hessian square-root source, continuum compiler, and projective record descent. Exact finite and operational constructions are distinguished from supplied continuum endpoints and apparatus clocks.

Retained The canonical binary law and second-moment descent remain closed on their declared domain. The concentration and persistence bounds below are unchanged. Measurement remains an ordinary physical interaction.

Open boundary A universal physical apparatus family, selected physical continuum repair-to-recorder coupling, and objective actualization are not inferred from the finite Hessian or the nonselective limit.

Version 2 Revision Note

Supersedes *Why the Born Rule and the Classical Limit Are the Same Problem: A Projection-Based Shadow Bridge in Modal Triplet Theory*, version 1.

Reason The earlier paper inferred squared-norm basin weights from projection, identified probability generation with classical concentration, and described both as a unified resolution. It did not state the measure and noncontextuality assumptions behind Gleason-style trace representation, and it did not distinguish record stability from selection of one outcome.

Resolution This version separates the quantum probability source, ordinary instrument dynamics, decoherence, concentration, record persistence, and one-history actualization. It imports Gleason-type representation at its proper conditional tier, records the exact current q79 binary-recorder result, and proves finite concentration and persistence bounds independently.

Retained result Born-compatible record weights and the classical limit can use the same outcome algebra and basin labels. Once a valid record law is known, concentration into one robust record gives a controlled deterministic approximation.

Remaining boundary MTT has not yet derived the required outcome law for every allowed apparatus context, a universal concentration regime, or an objective rule selecting one ontic history. Projection alone supplies none of these three results.

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1 The corrected relation

There are at least four distinct questions in a measurement process.

1. Which alternatives can become physical records?
2. What probabilities are assigned to those records?
3. When are interference terms between records operationally negligible?
4. When is one record so dominant and stable that a deterministic classical description is accurate?

A fifth question may be asked: why is one individual record actual rather than another? Standard operational quantum mechanics need not answer that question in order to predict recorded frequencies. Any ontic completion must answer it without confusing the answer with the probability law.

The old title compressed questions 2 and 4 into one problem. The corrected claim is narrower and more useful:

Born-compatible record law + concentration and stability $\implies$ controlled classical predictions

Each input has its own hypotheses and can fail independently.

2 Measurement as an ordinary physical instrument

Let $\mathcal{H}$ be a complex Hilbert space and ρ a density operator. A finite quantum instrument is a family

$$\{\mathcal{I}_i\}_{i \in I}$$

of completely positive, trace-nonincreasing maps such that $\sum_i \mathcal{I}_i$ is trace preserving. Its effects are

$$E_i = \mathcal{I}_i^*(\mathbf{1}), \quad E_i \geq 0, \quad \sum_i E_i = \mathbf{1}.$$

The operational outcome law and conditional post-measurement state are

$$p_i = \text{Tr}(\rho E_i), \quad \rho'_i = \frac{\mathcal{I}_i(\rho)}{p_i} \quad (p_i > 0). \quad (2.1)$$

Nothing in (2.1) makes measurement metaphysically privileged. An apparatus is a physical interaction that amplifies alternatives into records. The instrument formalism records the input-output statistics and state update of that interaction.

Suppose an upper model has measurable record regions B_i and an upper probability law $\mu_{\rho, \mathcal{I}}$ for the preparation and apparatus context. The exact compatibility equation is

$$\mu_{\rho, \mathcal{I}}(B_i) = \text{Tr}(\rho E_i) \quad \text{for every } i. \quad (2.2)$$

Writing down normalized basin weights does not prove (2.2). The source law, the record regions, and the equality to the trace weights all require independent construction.

3 What Gleason-type theorems establish

Gleason’s theorem starts with a probability measure on the closed subspaces, or equivalently projections, of a real or complex Hilbert space of dimension greater than two. Countable additivity on mutually orthogonal subspaces implies that the measure has the trace form

$$\mu(P) = \text{Tr}(\rho P)$$

for a positive trace-class operator ρ [1]. Extensions using positive-operator-valued measurements recover the trace form for generalized effects and can include two-dimensional systems [2].

These are representation theorems. Their assumptions already include a probability assignment satisfying strong consistency conditions. They show the form that such an assignment must take; they do not derive an upper physical measure from non-injective projection.

For an MTT application, the logical order is therefore:

$$\begin{array}{l} \text{selected preparation and record measure} \\ \text{+additivity or effect noncontextuality} \end{array} \implies \text{trace representation,}$$

followed by a separate proof of the basin–trace equality (2.2). This separation prevents a conditional reconstruction from being reported as a source theorem.

4 Current q79 Born status

4.1 The exact canonical domain

The current q79 program closes a specific operational domain. For the canonical binary one-anchor recorder:

- the finite Hilbert/state/observable data and reduced dynamics are fixed;
- the commuting nondemolition Fock output algebra supplies the record events;
- the selected normal state emits the stopped output measure;
- second-moment capture descent is exact; and
- no separate Born axiom, stochastic primitive, observed probability, or numerical fit is added on that domain.

Thus equation (2.2) has a selected operational realization for that binary apparatus context. This is stronger than the conditional Gleason-only status of the earlier paper. The following source chain explains why; it does not replace ordinary normal-state operational semantics with a pre-quantum probability theory.

4.2 The stopped output law and second-moment descent

Let $P, Q = I - P$ be the canonical orthogonal projectors and let $u = t/t_0$, $\gamma = \log 448$. The two-channel vacuum Fock coupling uses $L_P = \sqrt{\gamma}P$, $L_Q = \sqrt{\gamma}Q$. Its commuting nondemolition output algebra is the algebra of record events, and restriction of the selected normal state supplies its probability measure [4]. The no-count amplitude and its effect are different:

$$K_0(u) = e^{-\gamma u/2}I, \quad K_0(u)^* K_0(u) = e^{-\gamma u}I.$$

For a first count at s with label $a \in \{P, Q\}$, and for the ready event at horizon u , the stopped law and instrument are

$$\begin{aligned}\mu_\rho(ds, a) &= \gamma e^{-\gamma s} \text{Tr}(\rho P_a) ds, \quad 0 < s \leq u, \\ \mathcal{I}_{\text{ready}}^u(\rho) &= e^{-\gamma u} \rho, \\ \mathcal{I}_a^u(\rho) &= (1 - e^{-\gamma u}) P_a \rho P_a.\end{aligned}$$

The instrument sums to the dephasing channel, not to one selected outcome. Its normalization follows from $P + Q = I$. For the selected first carrier basis preparation, the conditional label weights are $1/3, 2/3$; at $u = 1$ the ready, P , and Q weights are $(1, 149, 298)/448$. These numbers are consequences of the declared source and clock, not fitted frequencies.

If two normalized preparation ensembles have the same second moment $\rho = \int |z\rangle\langle z| d\lambda(z)$, every stopped event and state-valued instrument output agrees. This is the exact capture descent. For example, the equal ensemble of a unit vector in each of P, Q and the equal ensemble of their normalized plus/minus superpositions have the same density matrix and recorder label weights $1/2, 1/2$. Their bare fourth moments $\int \|Pz\|^4 d\lambda$ are $1/2$ and $1/4$. The latter statistic is a counterexample to an unrestricted geometric basin claim, not a counterexample to this output measure: it is not the recorder's event effect.

4.3 Why the meter and its clock are part of the source

The minimal-action source distinguishes two constructions that should not be conflated [5]. A two-state defect meter can use $K_r = P + cQ$, $K_d = sQ$, where $c^2 + s^2 = 1$. A ready-plus-two-label meter instead uses $K_r = cI$, $K_P = sP$, $K_Q = sQ$, retaining information about both channels. Fresh three-state pointers with $c^2 = e^{-\gamma\Delta}$ reproduce the exact channel and stopped law on the time grid. Their Fock limit supplies the continuous recorder.

A fixed bounded Hamiltonian on a finite meter cannot generate an exact positive-rate Markov dephasing law from time zero in a product preparation: the reduced derivative at zero is Hamiltonian, while dissipative loss starts at second order. Repeated-interaction couplings with the singular $\Delta^{-1/2}$ scaling evade that finite-meter obstruction in the specified limit. This is a limit theorem, not an exact finite-bandwidth apparatus theorem. Nor does the unread channel identify its instrument: an informative meter and a state-independent phase-flip record can realize the same dephasing channel.

The Hessian-source refinement identifies what the repair action contributes [6]. With the finite defect normalization $D_{\text{fin}}^* D_{\text{fin}} = Q$,

$$S_{\text{fin}}(w) = \frac{\kappa}{4|G|} \sum_{g \in G} \|(I - \rho(g))w\|^2 = \frac{\kappa}{2} \|D_{\text{fin}} w\|^2,$$

the positive Hessian square root is $\sqrt{\kappa}Q$. It selects the zero/positive support and hence the binary projector pair. The declared minimal-disturbance meter, preparation-blind clock, and apparatus context then give the minimal recorder source. This does not add a new universal measurement axiom. It also does not identify the stiffness κ with an independently predicted physical detector rate.

For unequal channel rates the total hazard in a normalized pure state is $\gamma_P \|P\psi\|^2 + \gamma_Q \|Q\psi\|^2$. It is preparation-blind exactly when the two rates agree. Thus equal rates implement a stated clock condition, not a conclusion of the projector alone. Normalized unequal-angle instruments exist and explain why specifying only the repair Hessian would leave apparatus freedom. Pure repair energy is not the physical signed action, and neither cost descent nor this positive square root selects one ontic record.

4.4 The conditional continuum compiler

The continuum compiler begins with a *supplied* positive self-adjoint augmented Hessian, an invariant three-mode sector with kernel/support ranks one and two, and a selected isometry T intertwining that sector's P/Q with the finite projectors [7]. With the same clock, minimal Luders coupling, and no added Hamiltonian, intertwining the bounded jump and drift operators lifts to the Fock cocycle by uniqueness. Consequently the stopped measure and state-valued paths transport as well. An exact Fourier-conjugated three-mode example with Hessian eigenvalues 0, 2, 5 demonstrates the compiler; it is not selection of a physical continuum HYM Hessian.

The approximate version gives a useful consumer contract. For an isometry T with projector defect $\varepsilon = \|P_f T - T P_c\|$, the projector-weight error is at most ε , and the finite-horizon label-mass error is at most $(1 - e^{-\gamma u})\varepsilon$. The unnormalized projected state has trace-norm error at most $2\varepsilon + \varepsilon^2$. If both conditional weights are at least $m > 0$, a conservative normalized-state bound is

$$\|\rho'_{a,f} - T\rho'_{a,c}T^*\|_1 \leq \frac{2(2\varepsilon + \varepsilon^2)}{m}.$$

Without the weight floor, rare-event normalization need not be stable. These estimates require the declared isometry and do not control arbitrary nonisometric encoders or unproved cutoff tails.

The still missing physical inputs are the selected continuum endpoint, metric, Hessian, low sector, finite comparison map and tails, and apparatus clock. This is an internal geometric source problem, not a reopening of the already established canonical Cauchy quantum model or its binary law.

4.5 Records on projective overlaps and at long times

The projective descent result transports both probabilities and conditional states [8]. On overlaps, $\rho_j = U_{ij}\rho_i U_{ij}^*$ and $P_{a,j} = U_{ij}P_{a,i}U_{ij}^*$; the instrument intertwines under the same conjugation. Traces agree and central projective phases cancel. Thus one need not choose global vector phases to define physical record laws. This is a statement on the specified open constant-rank stratum, not an automatic extension through the example's harmonic rank jump from three to seven.

For the selected root preparation, the long-time nonselective state has nonzero eigenvalues $1/3, 2/3$, hence purity $5/9$. It is neither of the two selective states. This example connects directly to the next section: exact loss of cross-sector coherence and exact Born-compatible record weights need not make one record dominant. Projective consistency also does not decide which record is actual. No new role for an observer is required for either the ordinary instrument or its conditional state update.

4.6 Why an output law does not choose an ontology

The Locality companion contextualizes the frozen operational ontology non-entailment result [10, 9]. At the canonical checkpoint, a 448-atom probability space with record classes of sizes 1, 149, and 298 gives one label per atom and the same law as the unitary recorder. Attaching its ordinary conditional states also preserves the instrument, not just the three probabilities. A coactual completion can retain the same operational data. Hence those data select neither ontology. This is a context-specific countermodel, not a settings-independent global hidden-variable theory or a physical actualization source. It supports the distinction

between record probability, concentration, and actualization used in this paper without duplicating the source theorem.

4.7 The quantifier boundary

The canonical result does not yet prove:

- the same descent for every allowed apparatus and preparation context;
- controlled corrections for finite-bandwidth or non-Markov detectors;
- a pre-quantum probability semantics, if one is demanded;
- or objective selection of one ontic history.

The correct status is therefore tiered:

canonical q79 binary one-anchor output law:	exact,
universal physical apparatus family:	open,
objective single-history actualization:	open.

5 Decoherence is not concentration

Let $\mathcal{D}$ be dephasing in a pointer decomposition. If

$$\frac{1}{2}\|\rho - \mathcal{D}(\rho)\|_1 \leq \delta,$$

then every effect $0 \leq E \leq \mathbf{1}$ satisfies

$$|\mathrm{Tr}(\rho E) - \mathrm{Tr}(\mathcal{D}(\rho)E)| \leq \delta. \quad (5.1)$$

This is the operational meaning of approximate decoherence for the declared effect family: coherences alter probabilities by at most δ .

Equation (5.1) does not imply that one diagonal weight is near one. The state

$$\frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

is exactly decohered and maximally nonconcentrated on its two pointer records. Decoherence helps explain stable alternatives and suppression of interference [3]; a separate concentration estimate is needed for an approximately deterministic record.

6 The finite classical concentration theorem

Let I be a finite record set and let

$$p = (p_i)_{i \in I}$$

be a probability law. For a function $f : I \rightarrow \mathbb{R}$, define

$$\mathrm{osc}(f) = \max_{i \in I} f(i) - \min_{i \in I} f(i).$$

Theorem 6.1 (Classical concentration). *Suppose a record $i_\star$ satisfies*

$$p_{i_\star} \geq 1 - \varepsilon \quad (0 \leq \varepsilon \leq 1).$$

Then

$$d_{\text{TV}}(p, \delta_{i_\star}) = 1 - p_{i_\star} \leq \varepsilon, \quad (6.1)$$

$$\left| \sum_{i \in I} p_i f(i) - f(i_\star) \right| \leq \varepsilon \operatorname{osc}(f) \quad (6.2)$$

for every real function f on I .

Proof. Using the convention $d_{\text{TV}}(p, q) = \frac{1}{2} \sum_i |p_i - q_i|$,

$$d_{\text{TV}}(p, \delta_{i_\star}) = \frac{1}{2} \left(1 - p_{i_\star} + \sum_{i \neq i_\star} p_i \right) = 1 - p_{i_\star}.$$

Also,

$$\sum_i p_i f(i) - f(i_\star) = \sum_{i \neq i_\star} p_i (f(i) - f(i_\star)).$$

Taking absolute values and using $|f(i) - f(i_\star)| \leq \operatorname{osc}(f)$ gives (6.2). $\square$

This theorem gives an exact error budget for replacing the record law by the deterministic prediction $i_\star$. It does not say why $p_{i_\star}$ is large. That must follow from the preparation, dynamics, environment, control regime, or an MTT source theorem.

7 Record persistence

Concentration at one time is not enough. A classical record should persist. Let $X_0, X_1, \dots$ be a discrete-time record process on I . No Markov assumption is needed for the following conditional escape bound.

Theorem 7.1 (Finite-horizon persistence). *Assume*

$$\Pr(X_0 = i_\star) \geq 1 - \varepsilon$$

and, for every $k < n$,

$$\Pr(X_{k+1} \neq i_\star \mid X_0 = \dots = X_k = i_\star) \leq \eta.$$

Then

$$\Pr(X_0 = \dots = X_n = i_\star) \geq (1 - \varepsilon)(1 - \eta)^n \geq 1 - \varepsilon - n\eta. \quad (7.1)$$

Proof. The chain rule for conditional probabilities gives the first inequality. Bernoulli's inequality gives $(1 - \eta)^n \geq 1 - n\eta$. Finally,

$$(1 - \varepsilon)(1 - n\eta) \geq 1 - \varepsilon - n\eta.$$

$\square$

Equation (7.1) separates two classicality controls:

$$\varepsilon = \text{initial nonconcentration}, \quad \eta = \text{per-step record escape}.$$

A useful classical regime requires both to be small over the physical observation horizon.

8 How the two theorems fit together

Layer	Required object	What it establishes
Alternatives	instrument effects and record algebra	available records
Probability source	selected state/measure and capture map	the values p_i
Trace representation	additivity or effect noncontextuality	$p_i = \text{Tr}(\rho E_i)$
Decoherence	suppression of off-diagonal influence	classical mixture approximation
Concentration	$p_{i_\star} \geq 1 - \varepsilon$	deterministic prediction with error ε
Persistence	escape control η	stable record over a finite horizon
Actualization	interpretation or additional dynamics	why one history is individually actual

The same record labels and effects can appear in every row. That shared interface is the legitimate connection between the Born law and the classical limit. The logical arrows are not reversible. Concentration cannot derive the general Born weights, and a Born law need not be concentrated.

9 A precise MTT completion target

To extend the canonical q79 result into a general Born-and-classical theorem, MTT must supply a family indexed by physical contexts c and a classical regime parameter L :

$$(\rho_{c,L}, \{\mathcal{I}_{i,c,L}\}, \mu_{c,L}, \{B_{i,c,L}\}).$$

The required statements are:

1. **Source equality.**

$$\mu_{c,L}(B_{i,c,L}) = \text{Tr}(\rho_{c,L} E_{i,c,L})$$

for every allowed context and outcome, with stated errors for nonideal detectors.

2. **Concentration regime.** There is a selected $i_\star(c, L)$ and an explicit $\varepsilon(c, L) \rightarrow 0$ such that

$$\mu_{c,L}(B_{i_\star,c,L}) \geq 1 - \varepsilon(c, L).$$

3. **Persistence regime.** A derived escape bound $\eta(c, L)$ obeys

$$n(L)\eta(c, L) \rightarrow 0$$

on the intended observation horizon.

4. **Domain and interpretation.** The theorem states whether it is an operational record theorem only or also claims an ontic one-history law.

When these four rows are proved from the same selected source, the previous informal slogan can be replaced by a valid implication:

$$\begin{array}{l}
 \text{Born-compatible source law} \\
 \text{+decoherence} \\
 \text{+concentration} \\
 \text{+persistence}
 \end{array}
 \implies \text{controlled classical record theory.}$$

10 Consequences and non-consequences

The corrected separation yields several firm conclusions.

- Measurement can remain an ordinary physical interaction; no special conscious observer is needed in the mathematics.
- The current q79 binary recorder is a real advance over a purely conditional Gleason reconstruction.
- Classical behavior is quantitative: it comes with the three errors δ , ε , and $n\eta$ for decoherence, concentration, and persistence.
- Exact decoherence does not imply a deterministic outcome.
- A sharply concentrated law does not explain the general Born rule.
- Operational probabilities do not, by themselves, select one ontic history.
- Projection can define record equivalence classes, but it does not create their measure.

The phrase “same problem” should therefore be retired. The two problems are adjacent stages of one physical workflow and can be tested on one common instrument, but their proof obligations remain distinct.

11 Conclusion

Born probabilities and the classical limit share an outcome algebra, not a single theorem. A Born source theorem fixes the weights of records. Decoherence controls interference. Concentration makes one record approximately deterministic, and persistence keeps it stable. One-history actualization, if required, is another question.

MTT now closes more of this chain than version 1 reported correctly. The canonical q79 binary one-anchor Fock recorder has an exact stopped-output law and exact second-moment capture descent on its declared domain, without a fitted probability. The general apparatus family remains open. The classical concentration and persistence theorems proved here then state exactly what must be shown, after a valid record law exists, to obtain a controlled classical regime.

This separation is not a retreat from unification. It is the structure needed for a rigorous one: one source may eventually discharge several adjacent proof obligations, but none is counted as solved merely because the same record labels appear in all of them.

Computational Evidence and Reproducibility

The concentration and persistence bounds are proved in this paper. The five contextual sources cited in Section 4.2 and its following subsections are frozen at manifest commit `f141a20e`

a23c5c3f f19cc216 1c0e226e 29ade8a7. Their full source explanations and byte hashes were reviewed; packet flags are not independent theorem verification. The general apparatus and actualization boundaries remain open. The older strict-upgrade ledger below is historical provenance for a separate source boundary, not a current global completion count or a source of new outcome weights.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Historical 2/9 strict no-knob ledger snapshot; its count is not promoted to current authority.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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- [5] P. Nero, *Minimal q79 recorder action*, frozen contextual source (2026). https://github.com/PeterNero/mtt-results-repro/blob/f141a20ea23c5c3ff19cc2161c0e226e29ade8a7/release/results/q79_minimal_recorder_action/artifact.json.

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https://github.com/PeterNero/mtt-results-repro/blob/f141a20ea23c5c3ff19cc2161c0e226e29ade8a7/release/results/q79_hessian_recorder_source/artifact.json.
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